

Imperial College London

BSc/MSci EXAMINATION May 2018

This paper is also taken for the relevant Examination for the Associateship

PRINCIPLES OF INSTRUMENTATION

For 3rd/4th-Year Physics Students

Thursday, 24th May 2018: 10:00 to 12:00

The paper consists of two sections: A & B.

Section A contains one question, comprising of small parts. [40 marks total]

Section B contains four questions on selected parts of the course. [30 marks each]

Table of Laplace transformations is included at the end of the paper.

Candidates are required to:

*Answer **ALL** parts of Section A and **TWO** questions from Section B.*

Marks shown on this paper are indicative of those the Examiners anticipate assigning.

General Instructions

Complete the front cover of each of the THREE answer books provided.

If an electronic calculator is used, write its serial number at the top of the front cover of each answer book.

USE ONE ANSWER BOOK FOR EACH QUESTION.

Enter the number of each question attempted in the box on the front cover of its corresponding answer book.

Hand in THREE answer books even if they have not all been used.

You are reminded that Examiners attach great importance to legibility, accuracy and clarity of expression.

SECTION A

1. (i) A photomultiplier tube (PMT) has 12 dynodes and a photocathode with quantum efficiency of 30% for light with 175 nm wavelength. On average each dynode emits 2.6 **additional** electrons per incident electron.
- Calculate the total gain of the dynode system.
 - Calculate the average total charge created per individual photon of 175 nm wavelength entering the tube.
 - Estimate the current at the output of the PMT created by a single photoelectron, assuming that average electron transit time spread through the PMT is 10 ns (Full Width at Half Maximum - FWHM).

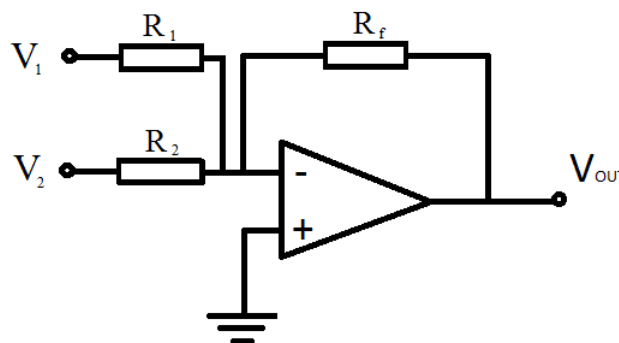
[9 marks]

- (ii) The same PMT is connected to the data acquisition electronics via a co-axial cable with a capacitance per metre $C' = 95.8 \text{ pF m}^{-1}$ and inductance per metre $L' = 0.2395 \text{ } \mu\text{H m}^{-1}$.

- Calculate the characteristic impedance of the cable and explain its physical meaning.
- Calculate the time delay caused by the cable of length 12.8 m.
- Estimate the voltage on the cable created by a single photoelectron emission.

[10 marks]

- (iii) (a) Briefly state the properties of the ideal operational amplifier and give the two 'golden-rules' used to simplify the analysis of circuits containing them.
- (b) The diagram below shows a circuit with an op-amp which is assumed to be ideal. Calculate the output voltage V_{out} as a function of the input voltages V_1 and V_2 , and values of the resistances of the resistors R_1 , R_2 and R_f .



(c) With $R_1 = 10 \text{ k}\Omega$, $R_2 = 20 \text{ k}\Omega$ and $R_f = 20 \text{ k}\Omega$, sketch V_{out} for

$$V_1(t) = \begin{cases} 0 \text{ V} & t < -1 \text{ s} \\ -0.5 \text{ V} & -1 \text{ s} \leq t \leq 0 \text{ s} \\ 1 \text{ V} & 0 \text{ s} \leq t \leq 1 \text{ s} \\ 0 \text{ V} & t > 1 \text{ s} \end{cases}$$

and

$$V_2(t) = \begin{cases} 0 \text{ V} & t < -1 \text{ s} \\ 1 \text{ V} & -1 \text{ s} \leq t \leq 0 \text{ s} \\ 2 \text{ V} & 0 \text{ s} \leq t \leq 1 \text{ s} \\ 0 \text{ V} & t > 1 \text{ s} \end{cases}$$

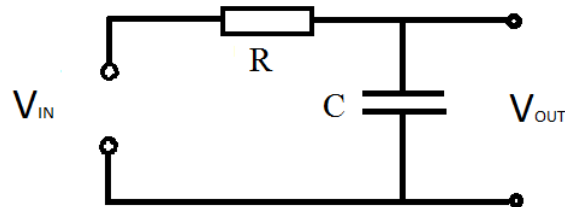
(d) Briefly state the properties of typical 'real-world' operational amplifiers.

[21 marks]

[Total 40 marks]

SECTION B

2. (i) Consider the RC circuit shown in the diagram below. The input voltage V_{in} is the step function ($V_{in}(t) = 0$ for $t < 0$ and $V_{in} = V$ for $t \geq 0$) and the capacitor is initially charged to voltage $V/2$. Consider the behaviour of the circuit for $t > 0$.



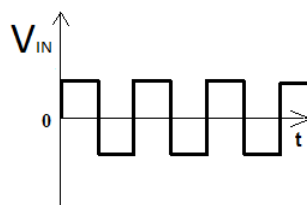
- (a) Write the differential equation governing the current flowing in the circuit.
- (b) Derive the expression for the current flowing in the circuit as a function of time.
- (c) Calculate and sketch the output voltage V_{out} .

[11 marks]

- (ii) Now consider the same circuit as shown in the diagram above, but assume the input voltage is provided by the AC voltage source.
 - (a) Assuming $R=4\text{ k}\Omega$ and $C=0.25\text{ }\mu\text{F}$ calculate the cut-off frequency of the circuit and make a sketch of the Bode plot.
 - (b) Explain the behaviour of this circuit as a filter and propose methods to improve its behaviour in terms of steepness of the frequency roll-off, flatness of the pass-band and sharpness of the knee.

[7 marks]

- (iii) Sketch the output voltage V_{out} of this circuit connected to the input source producing an ideal rectangular voltage signal, shown in the figure below, with the frequency much smaller than the cut-off frequency of the circuit. Comment briefly on its shape.



[4 marks]

- (iv) (a) Explain briefly properties of the flicker noise in measurements and describe methods of its mitigation.
- (b) The signal from the instrument affected by the flicker noise is measured over a period of five minutes and the flicker noise power is measured with a bandwidth of 5 kHz. After what period of time would the noise power be expected to increase by 25 %?

[8 marks]

[Total 30 marks]

3. (i) (a) Define the Laplace transform $F(s)$ of the continuous function $f(t)$.
 (b) Prove the transform's integral property:

$$\mathcal{L}\left\{\int_0^t f(\tau)d\tau\right\} = F(s)/s$$

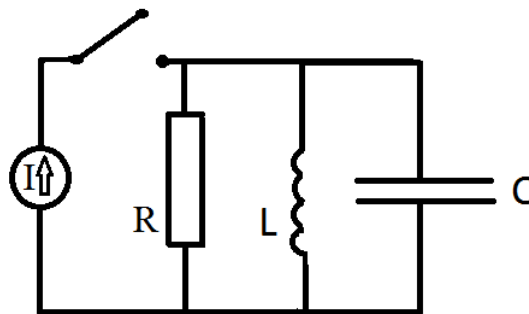
Hint: You may use:

$$\mathcal{L}\left\{\frac{df}{dt}\right\} = sF(s) - f(0^+)$$

where $f(0^+)$ is the initial condition for $f(t)$.

[7 marks]

- (ii) Consider the parallel LRC circuit shown below consisting of an ideal current source of 2 A, inductance of 0.5 H, capacitance of 0.03 F, resistance of $10\ \Omega$ and a switch. Assume that when the switch is turned on at time $t = 0$, the capacitor is not charged.



- Write the equation describing the conservation of current flowing in the circuit in terms of the partial currents flowing through the resistor, the inductor and the capacitor.
- Write the differential equation describing the time evolution of current flowing through the inductor.
- Take the Laplace transform of the equation describing the current flowing in the inductor and write its corresponding s-domain representation.
- By performing the inverse Laplace transform, find an expression for the current flowing through the inductor for time $t > 0$ after closing the switch. **Hint:** You may use the table of standard Laplace Transforms at the end of this paper.
- Calculate voltage across the capacitor $V(t)$ after closing the switch and sketch its behaviour.

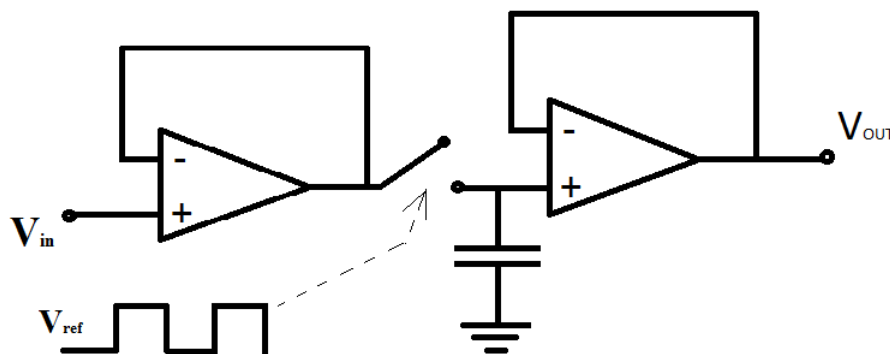
[23 marks]

[Total 30 marks]

4. (i) (a) State the Nyquist theorem and explain the meaning of the Nyquist frequency and the aliasing effect.
- (b) A signal analyser, which operates at the sampling rate of 1 GHz, is used to study four separate and distinct sinusoidal signals of frequency 360 MHz, 470 MHz, 530 MHz and 1.2 GHz. Decide which of these signals can be properly sampled, explain why and what are the alias frequencies for those, that cannot be?

[8 marks]

- (ii) In the circuit given below, the switch opens on a rising edge of V_{ref} and closes on a falling edge. The capacitor has a very small value and the signal V_{in} comes from an ideal voltage source. Sketch the output voltage assuming V_{in} to be sinusoidal voltage with a time period four times longer than the time period of V_{ref} . Comment on the applicability of this circuit in signal processing.



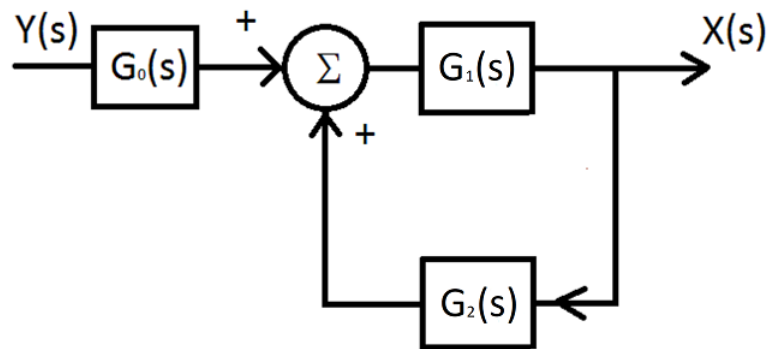
[5 marks]

- (iii) A 3-bit Analogue to Digital Converter (ADC) is used to digitise a positive voltage signal in the range 0-4 V.
- (a) Determine the resolution of the digitisation process.
- (b) What is the dynamic range, in dB, for the digitised signal?
- (c) Give the maximum quantisation error and the RMS quantisation noise added.
- (d) Sketch the circuit required for the 3-bit flash ADC type.
- (e) Comment on advantages and disadvantages of the flash ADC type.

[17 marks]

[Total 30 marks]

5. The system shown below contains five functional blocks and contains a feedback loop.



- (i) (a) Calculate the system transfer function $G(s)$ in terms of $G_0(s)$, $G_1(s)$ and $G_2(s)$.
- (b) $G_0(s)$ has a time-domain impulse response of $2te^{-t}$. Find its s-domain representation.
- (c) Assuming $G_1(s) = \frac{1}{s^2+3s+2}$, find its time-domain impulse response.
- (d) $G_2(s)$ is the ideal amplifier with gain k , where k represents a real constant. What is its s-domain representation?
- (e) Find the overall transfer function in terms of s .

[15 marks]

- (ii) (a) Determine the range of k values, in which the system is stable. Explain your answer.
- (b) For which values of k would the circuit oscillate?
- (c) Determine the system response to the unit step input in the time-domain assuming $k = 0.75$.

[15 marks]

Hint: You may use the table of standard Laplace Transforms at the end of this paper. [Total 30 marks]

Table of Laplace Transforms

$f(t)$	$F(s) = \mathcal{L}\{f(t)\}$
$\delta(t)$	1
$u(t)$	$\frac{1}{s}$
t^n for $n > 0$	$\frac{n!}{s^{(n+1)}}$
e^{-at}	$\frac{1}{s+a}$
te^{-at}	$\frac{1}{(s+a)^2}$
$\frac{1}{b-a}(e^{-at} - e^{-bt})$	$\frac{1}{(s+a)(s+b)}$
$\sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$
$\cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$
$e^{-at} \sin(\omega t)$	$\frac{\omega}{(s+a)^2 + \omega^2}$
$e^{-at} \cos(\omega t)$	$\frac{s+a}{(s+a)^2 + \omega^2}$